

# Hello! 🖐️



**Pieter De Vis**



**Sam Hunt**



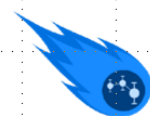
**Maddie Stedman**



**Samantha Malone**



**Joe Riordan**



## Today's Exercises



[comet-toolkit.org/user-  
guide/training/vhroda](https://comet-toolkit.org/user-guide/training/vhroda)



# Today's Agenda

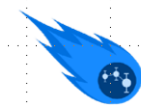


## 1. Presentation: CoMet Toolkit Introduction

- ☐ *What is CoMet?*
- ☐ *Uncertainty 101*
- ☐ *Tools Intro*

## 2. Exercises

- ☐  **punpy** basics with in-situ type data
- ☐  **obsarray** basics with EO type data



### Today's Exercises



[comet-toolkit.org/user-guide/training/vhroda](https://comet-toolkit.org/user-guide/training/vhroda)



# The CoMet Toolkit – Uncertainties made easy

Pieter De Vis, Sam Hunt, Maddie Stedman, Samantha Malone, Joe  
Riordan  
*National Physical Laboratory*

VHRODA hands-on tutorial - 19/11/2025

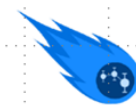


# Presentation Outline



## 1. CoMet Toolkit Introduction

- ☐ *Motivation & Overview*
- ☐ *Uncertainties 101*
- ☐ *CoMet Packages*
  -  **punpy**
  -  **obsarray**
- ☐ *Application examples*



### Today's Exercises

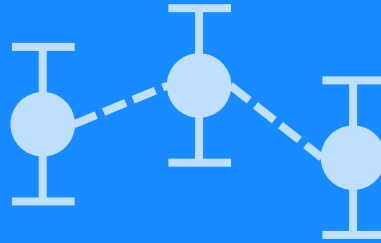


[comet-toolkit.org/user-guide/training/vhroda](https://comet-toolkit.org/user-guide/training/vhroda)

# CoMet Toolkit



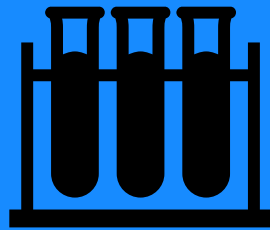
Python  
Tools



Uncertainty  
Handling



Open  
Source



Tested

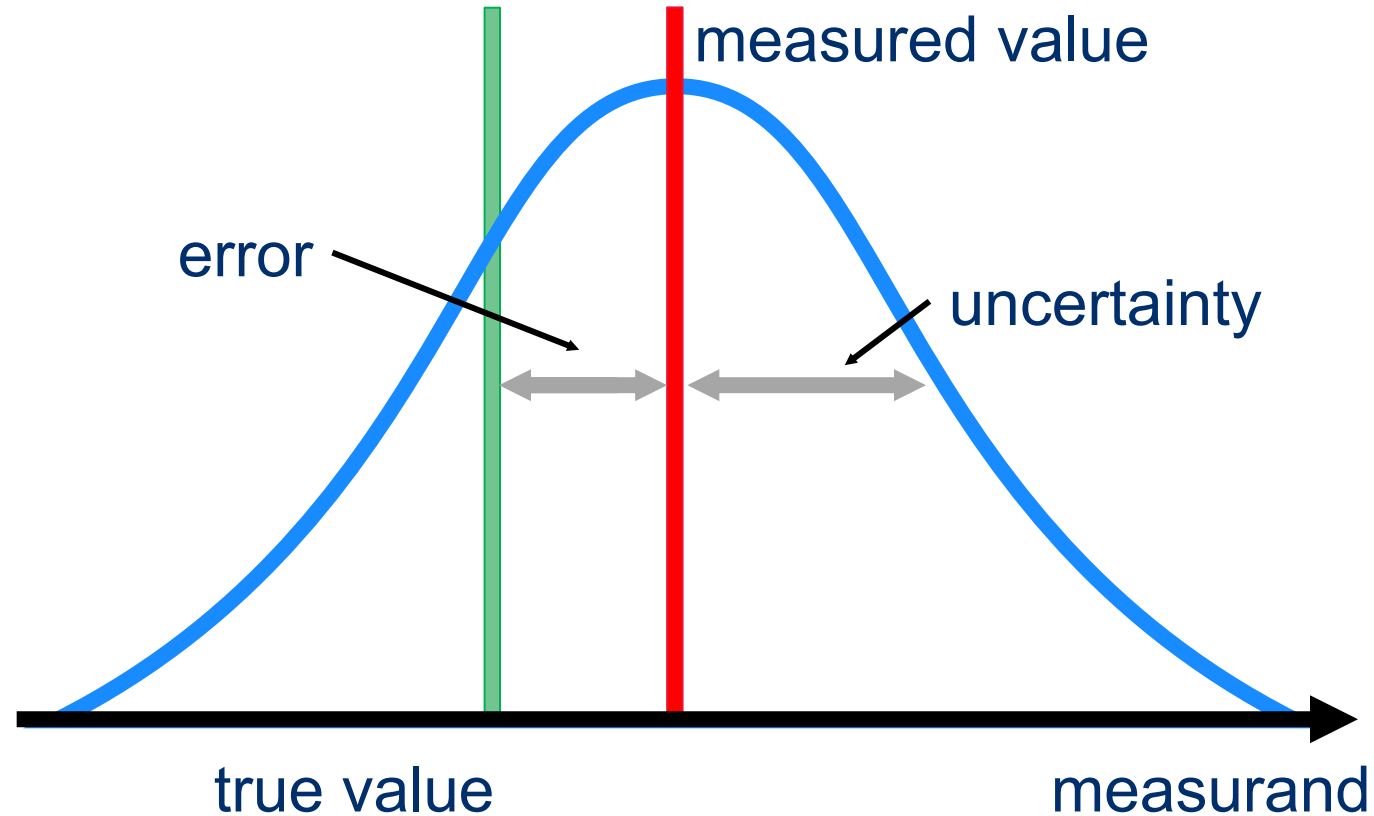


Applied

IDEAS-QA4EO



# Uncertainties 101





# Error Correlation



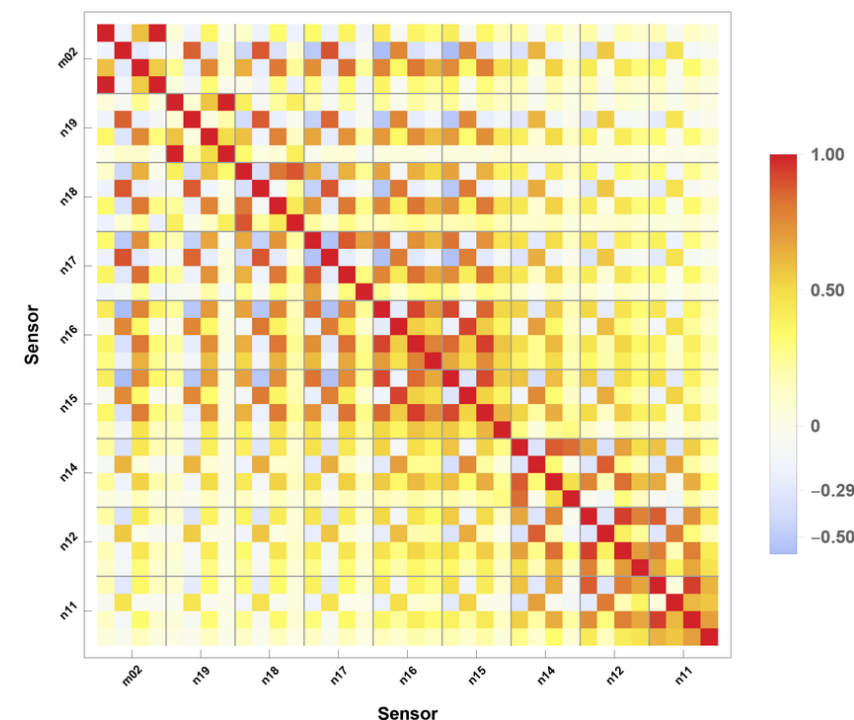
## What is Error Correlation?

Errors in a dataset are **not always independent** — e.g., when errors in pixels, bands instruments, or time steps are systematically related.



## Why does it Matter?

- ☐ Bias uncertainties persist – averaging doesn't help!
- ☐ Band ratios might be off – affecting retrieval uncertainty
- ☐ Misleading confidence in trends



*Error correlation matrix from  
Giering et al. 2019*



# Error Correlation



## What is Error Covariance?

Combines error correlation and uncertainty

$$S = U R U^T$$



## Random, systematic and structured uncertainty

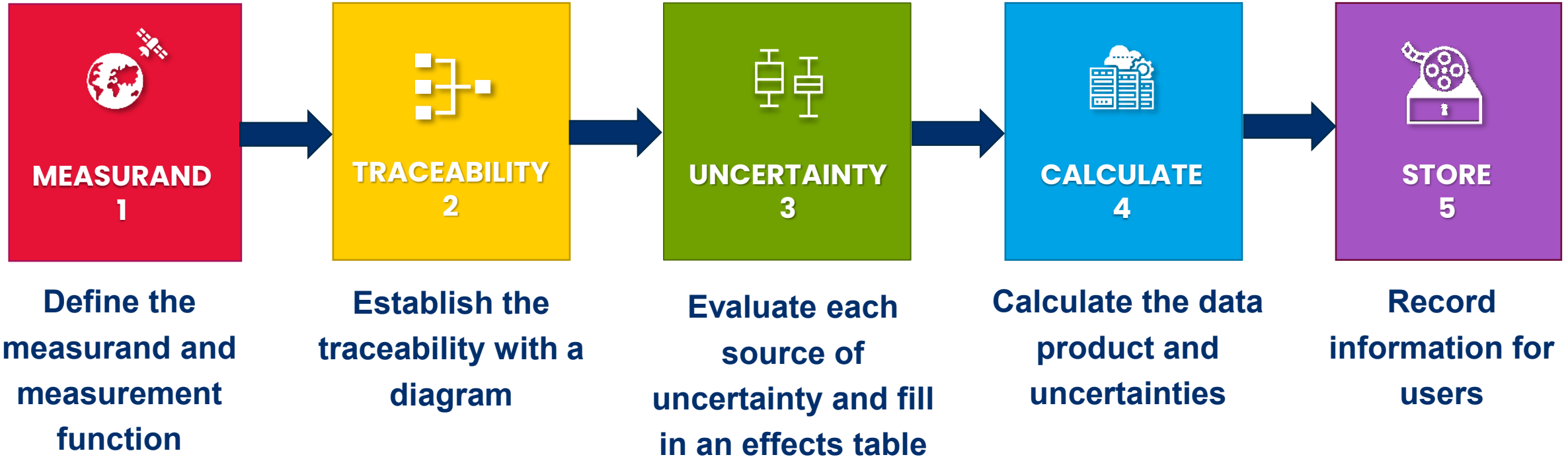
- ☐ It is the errors that are correlated, not the uncertainty values
- ☐ Random means completely uncorrelated, i.e. error correlation is identity matrix
- ☐ Systematic means fully correlated, i.e. error correlation filled with 1's



# A Metrological Approach



Uncertainties are evaluated and expressed following [QA4EO Five Steps](#), a framework which employs the principles of metrology.



# CoMet Toolkit



 **punpy**

Propagation UNcertainties in Python

 **obsarray**

Handling uncertainty and error-covariance in datasets

 **comet\_maths**

CoMet mathematical algorithms and interpolation tools

 **UNC Specification**

Uncertainty metadata naming conventions

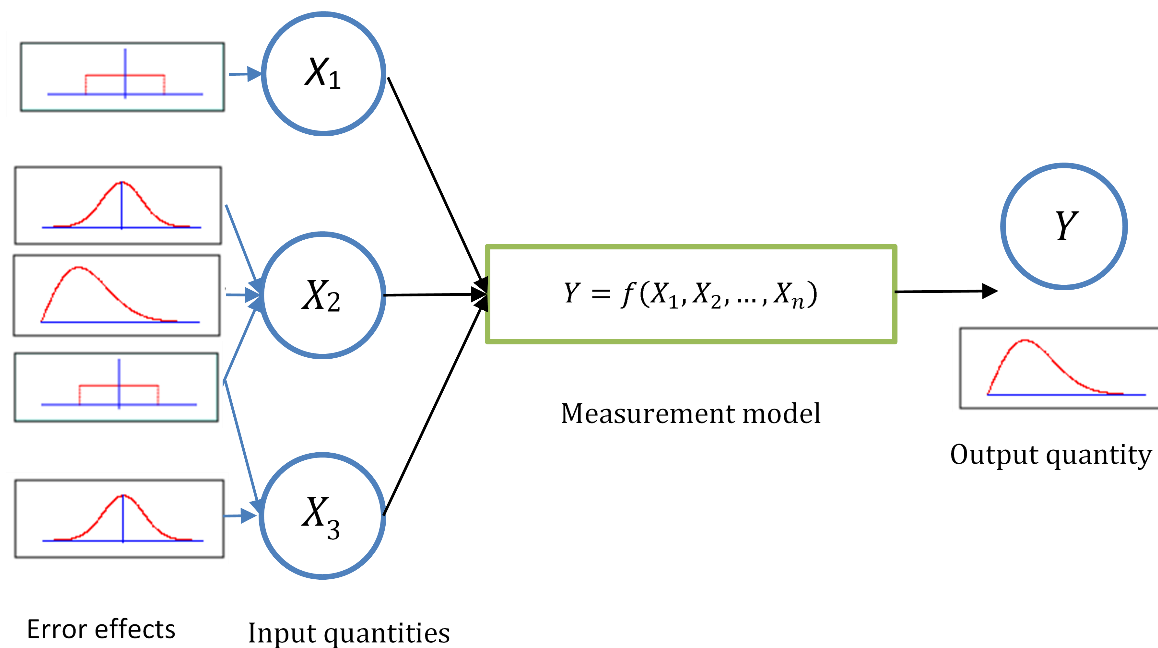
# CoMet Packages:



 punpy

# Propagating Uncertainties with punpy

- ❑ Python module for propagating **random**, **systematic** and **structured uncertainties** through any Python measurement function
- ❑ Flexible in terms of the **specified correlations** along given dimensions or between input quantities
- ❑ **Monte Carlo** and **Law of Propagation of uncertainties** methods available



# Punpy as a Standalone Tool



## □ Simple user **interface**:

1. Import punpy
2. Define measurement function
3. Create MC or LPU object
4. Propagate uncertainties

```
import punpy
prop=punpy.MCPropagation(10000)
unc_measurand=prop.propagate_random(measurement_func,
                                     [input_qt1,input_qt2],[unc_qt1,unc_qt2])
```

□ **Measurement function** are defined as python functions that take arrays as input quantities and return an array as measurand

□ Many optional **keywords** for flexible functionality

- *return\_corr*
- *Corr\_between*
- *Repeat\_dims*
- *Parallel\_cores*
- *Output\_vars*
- ...

# Punpy with digital effects tables

- ❑ **punpy** interfaces with **obsarray** to make uncertainty propagation as efficient and easy to use as possible
- ❑ **propagate\_ds()** function returns an **obsarray** dataset with combined random, systematic and structured uncertainties on measurand

```
from punpy import MeasurementFunction

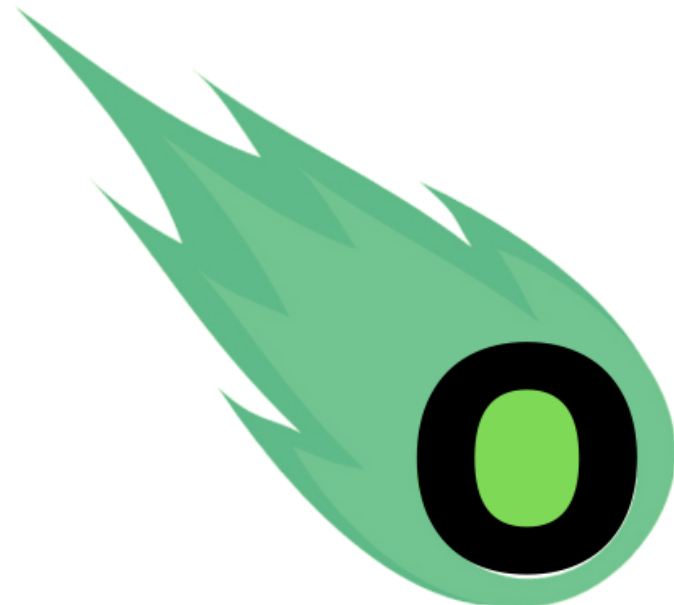
# Define your measurement function inside a subclass of MeasurementFunction
class IdealGasLaw(MeasurementFunction):
    def meas_function(self, pres, temp, n):
        return (n * temp * 8.134) / pres

# create object of the measurement function class and specify the variable names
gl = IdealGasLaw(["pressure", "temperature", "n_moles"], "volume", yunit="m^3")

# propagate uncertainties on the input quantities in ds to measurand in ds_y
ds_y = gl.propagate_ds(ds)
```

 CoMet Packages:



 obsarray



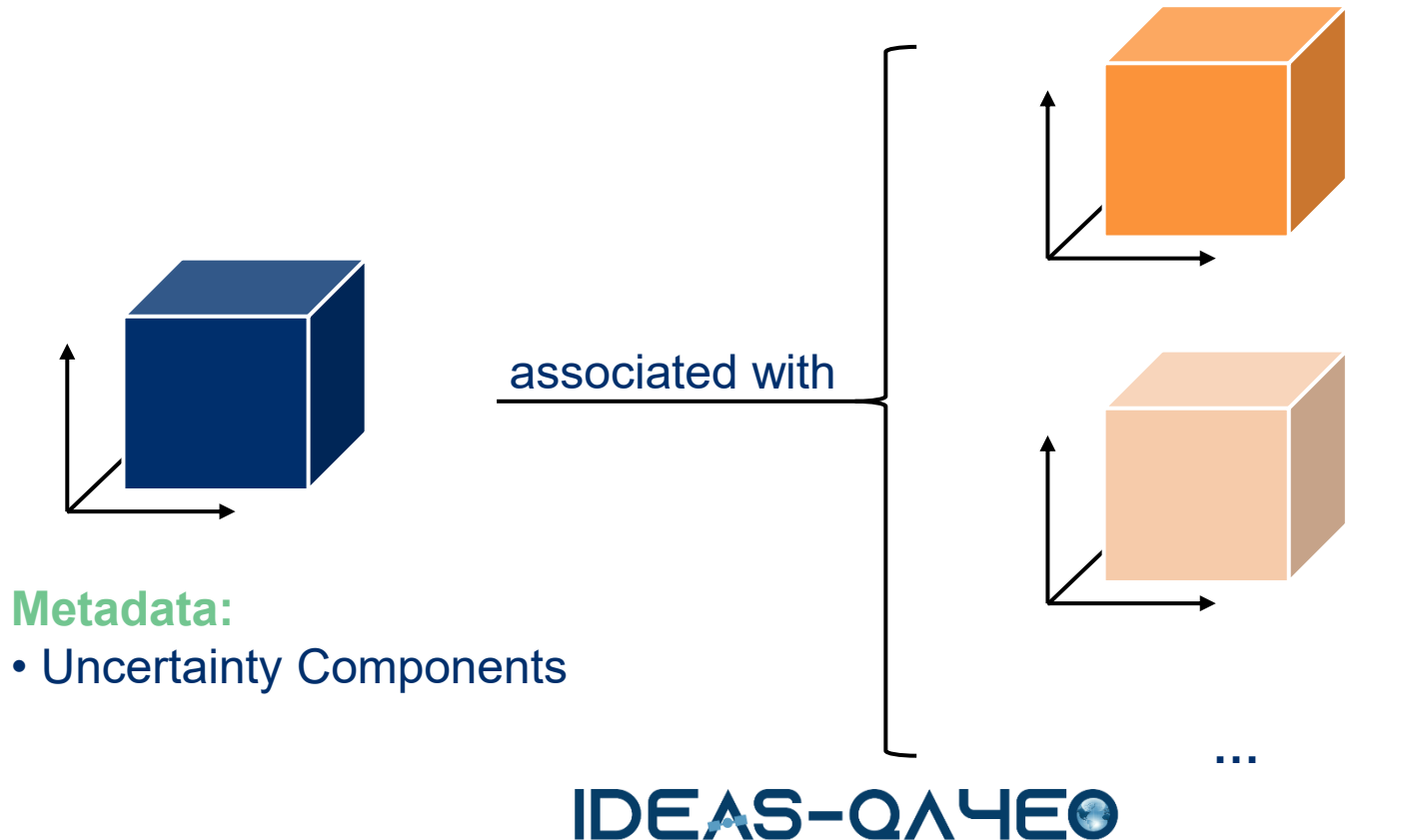
# UNC Specification



## Uncertainty Variable Metadata

Observation Variables

Uncertainty Variables



Metadata:

- **PDF Shape** (gaussian, ...)
- **Units** (abs. or rel.)
- **Error-Correlation...**

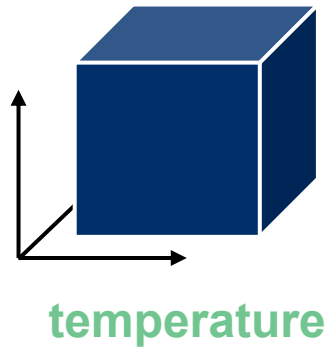


# UNC Specification



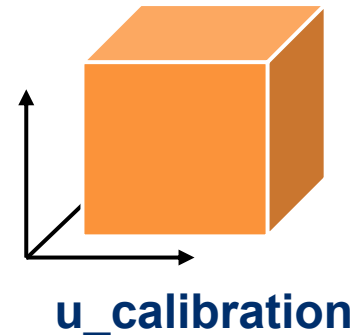
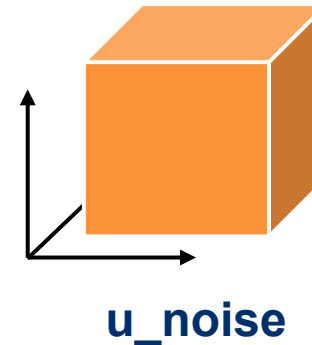
## Example – Temperature Dataset

### Observation Variables



associated with

### Uncertainty Variables



#### Metadata:

- PDF Shape – “gaussian”
- Units – %
- Error-Correlation:
  - All dims - Random

#### Metadata:

- PDF Shape – “rectangular”
- Units – “K”
- Error-Correlation:
  - **x, y** – systematic
  - **time** – defined by matrix



# UNC Specification



## Example – Temperature Dataset

```
variables:
  float temperature(time, lat, lon);
    temperature:unc_comps=["u_calibration", "u_noise"];
    temperature:units="K"
  float u_calibration(time, lat, lon);
    u_calibration:units="K";
    u_calibration:pdf_shape="rectangular";
    u_calibration:err_corr_dim1_name=["lat", "lon"];
    u_calibration:err_corr_dim1_form="systematic";
    u_calibration:err_corr_dim2_name="time";
    u_calibration:err_corr_dim2_form="err_corr_matrix";
    u_calibration:err_corr_dim2_params=["err_corr_calibration_time"];
  float u_noise(time, lat, lon);
    u_calibration:err_corr_dim1_name=["time", "lat", "lon"];
    u_calibration:err_corr_dim1_form="random";
  float err_corr_calibration_time(time, time);
```



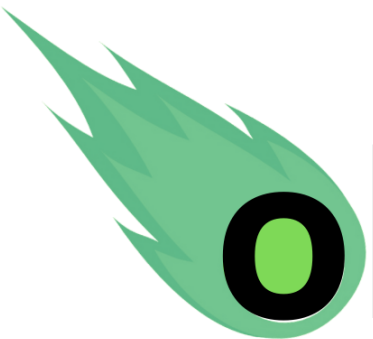
## Measurement data handling in Python

- ❑ **obsarray** is an extension to xarray to support defining, storing and interfacing with measurement data – using the UNC specification.
- ❑ Also has functionality for defining flags following **CF Conventions**.
- ❑ It is designed to work well with netCDF files and for the **Earth Observation** community.

**Plugs straight into punpy for propagation through measurement functions!**

# Comet Application Examples

 punpy

 obsarray

 comet\_maths

# CoMet Toolkit in Action



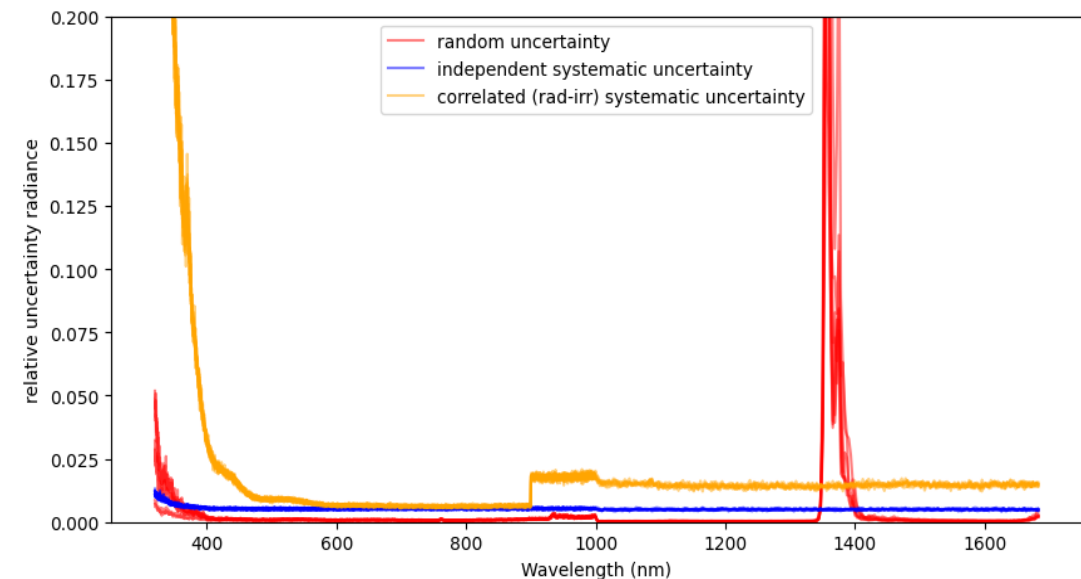
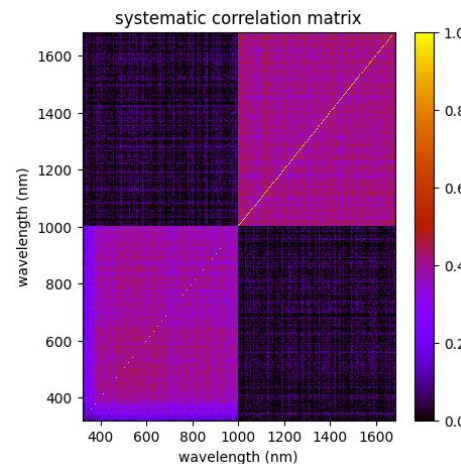
- ❑ Validated against **NIST** uncertainty engine:

[https://colab.research.google.com/github/comet-toolkit/comet\\_training/blob/main/NIST\\_example.ipynb](https://colab.research.google.com/github/comet-toolkit/comet_training/blob/main/NIST_example.ipynb)

- ❑ *CoMet* is used in various other projects, such as **QA4EO**, **HYPERNETS**, **CHIME L2**, **FLEX** validation, **TRUTHS** science studies, **LIME**, **FRM4SOC**, **RPV4PICS**  
see <https://www.comet-toolkit.org/projects/>

- ❑ **Example** from *hypernets\_processor* :

*Hypernets is an automated network of in-situ instruments measuring reflectance for L2 satellite validation*





# Exercises



□ Please go to [www.comet-toolkit.org/user-guide/training/](http://www.comet-toolkit.org/user-guide/training/)



## Today's Exercises



*comet-toolkit.org/user-  
guide/training/vhroda*

**IDEAS-QA4EO**



# CoMet Release



❑ V1.0 of Comet toolkit has been released as **open source** toolkit:

- [www.comet-toolkit.org](http://www.comet-toolkit.org)
- [github.com/comet-toolkit](https://github.com/comet-toolkit)



❑ Accompanied by training material (**Jupyter** notebooks hosted on google colab):

- [www.comet-toolkit.org/examples](http://www.comet-toolkit.org/examples)

❑ Documentation & ATBD for individual tools:

- [obsarray.readthedocs.io/en/latest/](http://obsarray.readthedocs.io/en/latest/)
- [punpy.readthedocs.io/en/latest/](http://punpy.readthedocs.io/en/latest/)
- [comet-maths.readthedocs.io/en/latest/](http://comet-maths.readthedocs.io/en/latest/)



# Outlook



- ❑ **Current release** will be presented De Vis & Hunt (in prep)
  
- ❑ Looking to continue to expand the use cases the developed tools
  - Aiming to enable uncertainty propagation through **any python measurement function**
  -  Please get in touch if you are interested!
  
- ❑ This has been our first step into this way of working, **many more ideas in a roadmap**
  - e.g. retrieval tool/optimisation, Look-up tables
  - MetEOR (e.g. Cal/Val pipeline, BRDF tool, ...)





# Summary



- ❑ The **CoMet toolkit** is an open-source software project to develop Python tools for the handling of error-covariance information in the analysis of measurement data
- ❑ This toolkit is based on **robust metrology**, and makes dealing with complexities of uncertainties much easier
- ❑ Includes **obsarray**, **punpy** & **comet\_maths** as initial offering, to be extended
- ❑ These tools are already being used operationally in various projects (e.g. Hypernets)